

AGENTS.md — Agent Execution Guidelines & Operating Invariants

Scope & Target Audience

This document governs all autonomous AI agents (Claude Code CLI via ECC, Prime Agent RLM loops, and background maintenance daemons) executing tools or applying skills within [skills-and-capabilities/](#).

Core Operating Invariants

1. **Deterministic Exit Codes:** Every Python script in this repository must return exit code `0` on success and non-zero (`1` or `2`) on failure. Agents must verify `$? == 0` before proceeding to downstream steps.
 2. **Read-Only Inspection:** Ingestion and parsing tools (`graph_builder.py`, `markor_sweeper.py`, `graph_sync.py`) must operate in read-only mode across target directories. Never mutate source code during an analysis pass.
 3. **Bounded Context Windows:** When reporting results to human operators, never dump full raw JSON dependency trees or huge bundles. Always use `--format md` or pipe through markdown summary formatters to conserve context tokens.
 4. **No Unaudited Dependencies:** All Python utilities in this repository rely strictly on Python 3 standard libraries (`ast`, `re`, `json`, `hashlib`, `pathlib`, `argparse`). Do not introduce external pip dependencies (`networkx`, `pandas`, `requests`) unless explicitly authorized in `UNRESOLVED.md`.
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Skill Application Protocol (The Matt Pocock Sequence)

When executing feature buildouts or complex refactors, agents must chain the prompt skills in this strict sequence:

[grill-me] —> [spec] —> [ticket] —> [tdd] —> [code-review]

1. **grill-me:** Interrogate ambiguous user prompts; extract missing constraints.
2. **spec:** Generate formal architectural specifications with W5+H identity and invariants.
3. **ticket:** Deconstruct specifications into atomic work units (< 150 LOC, < 3 files).
4. **tdd:** Author failing unit tests before touching implementation code.

5. **code-review**: Run `graph_builder.py` and verify AST stability before committing changes.